

Prepared by: Abrar Ahmed

IoT Hardware and Systems

Q.1 Which one of the following is not a temperature sensor?	
A. Thermoradiator	B. Thermistor
C. Resistor Temperature Detectors	D. Thermocouples
Correct Answer: A	
Reference: Book name: IoT Hardware and Systems, Page no 26	

Q.2 When was the Raspberry Pi launched?	
A. 2010	B. 2011
C. 2013	D. 2012
Correct Answer: D	
Reference: Book name: IoT Hardware and Systems, Page no 61	

Q.3 How many GPIO pins are there on Raspberry Pi 1?	
A. 26	B. 40
C. 30	D. 36
Correct Answer: 40	
Reference: Book name: IoT Hardware and Systems, Page no 90	

Q.4 What is the default password of Raspberry Pi?	
A. Password	B. Raspberry
C. Root	D. Pi
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 66	

Q.5 Which one of the following categories of processors is found in embedded systems?	
A. ASSP	B. ASIP
C. GPP	D. All of these
Correct Answer: D	
Reference: Book name: IoT Hardware and Systems, Page no 13	

Q.6 In Von Neumann architecture, what operation can a PC do at a time?	
A. Fetch an Instruction	B. Perform Read/Write
C. Both of them can be done but not together	D. Both of them can be done together
Correct Answer: C	
Reference: Book name: IoT Hardware and Systems, Page no 15	

Q.7 Which stage in IoT architecture consists of Data Acquisition Systems?	
A. Stage 1	B. Stage 2
C. Stage 3	D. Stage 4
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 23	

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Q.8 Which one of the following sensors detects ionic concentrations in water solutions?	
A. Chlorine Residual Sensor	B. Ph Sensor
C. Turbidity Sensor	D. Conductivity Sensor
Correct Answer: D	
Reference: Book name: IoT Hardware and Systems, Page no 27	

Q.9 Match the following devices with their respective functions				
S NO.	Left Column	S NO.	Right Column	Correct Answer
A1	A-D convertor	B1	Compares the output from the D-A converter	B1 B2 B3 B4
A2	D-A converter	B2	Customizes the system according to the requirements	B1 B2 B3 B4
A3	ASIC	B3	Converts analog signal sent by sensor into a digital signal	B1 B2 B3 B4
A4	Actuator	B4	Converts the digital data fed by the processor to analog data	B1 B2 B3 B4
Reference: Book name: IoT Hardware and Systems, Page no 12				

Q.10 You need to start a machine learning and artificial intelligence project over Raspberry Pi. The goal is to choose a framework, which is most widely known and is easily deployable on Pi for MI/AI. Which two frameworks would you choose? [Multiple Choice]	
A. Keras	B. Tensor flow
C. Open CV	D. Torch
Correct Answer: B, C	
Reference: Book name: IoT Hardware and Systems, Page no 63	

Q.11 Which one of the following commands in boot/config.txt forces the Pi to send display to HDMI?	
A. force_hdmi=1	B. hdmi_force_hotplug=1
C. hdmi_force=1	D. hdmi_display=1
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 112	

Q.12 How many components are there in an embedded system?	
A. 1	B. 2
C. 3	D. 4
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 16	

Q.13 What is the Harvard architecture named after?	
A. Harvard University	B. Harvard Mark 1 computer
C. Prof. John Baylish Harvard	D. Harvard Research labs
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 16	

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Q.14 What conversion does an A/D Convertor do?	
A. AC to DC	B. Asynchronous to Digital
C. Analog Values to Data Values	D. Analog to Digital
Correct Answer: D	
Reference: Book name: IoT Hardware and Systems, Page no 12	

Q.15 How many stages are there in IoT architecture?	
A. 3	B. 4
C. 5	D. 6
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 22	

Q.16 Which one of the following issues when a broadcasted message with stronger transmission power pretends that the HELLO message is coming from the base station?	
A. DoS Attack	B. Hello Flood Attack
C. Hello Theft Attack	D. Initiation Attack
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 56	

Q.17 Which one of the following is not an optical sensor?	
A. Pyrometer	B. Turbidity sensor
C. Photodetector	D. Proximity
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 31	

Q.18 Identify the option that is based on XML.	
A. MQTT	B. HTTP
C. XMPP	D. CoAP
Correct Answer: C	
Reference: Book name: IoT Hardware and Systems, Page no 42	

Q.19 Which of the following are examples of IoT data protocols? [Multiple Choice]	
A. CoAP	B. MQTT
C. HTTP	D. XMPP
Correct Answer: A, B, D	
Reference: Book name: IoT Hardware and Systems, Page no 38	

Q.20 In MQTT, what handles concurrently connected clients?	
A. Integrator	B. Complier
C. API	D. Broker
Correct Answer: D	
Reference: Book name: IoT Hardware and Systems, Page no 42	

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Q.21 In which of the following topologies do all nodes connect to a central node for data exchange?	
A. Tree	B. Star
C. Mesh	D. Bus
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 52	

Q.22 Sensor nodes in a sensor field collect data and route it to the _____	
A. BaseNode	B. Sink Node
C. Primary Node	D. Server Node
Correct Answer: B	
Reference: Book name: IoT Hardware and Systems, Page no 53	

Q23: Which one of the following options is an internet-utility protocol used in restrictive device and network environments?	
A. MQTT	B. HTTP
C. XMPP	D. CoAP
Correct Answer: D	
Reference: Book name: IoT Hardware And Systems, page no 40, Heading 3.3	

Q24: Which one of the following options is not a type of operator in Python?	
A. logical	B. identity
C. relational	D. membership
Correct Answer: C	
Reference: Book name: IoT Hardware And Systems, page no 128, Heading 9.3.3	

Q25: Which of the following option does not evaluate to TRUE in Python?	
A. Non-zero value	B. Non-null value
C. yes	D. 1
Correct Answer: C	
Reference: Book name: IoT Hardware And Systems, page no 128, Heading 9.3.4	

Q26: In simple terms, what is virtual sensor?	
A. Emulation of a sensor	B. A device that behaves like a sensor but is actually not a sensor
C. Group of sensors	D. Unregistered sensors
Correct Answer: A	
Reference: Book name: IoT Hardware And Systems, Direct reference not available	

Q27: Which one of the following options is a machine to machine (M2M) platform as a service (PaaS) that manages communication between remote device assets and enterprise applications?	
A. Nimbits	B. Thing Speak
C. iDigi	D. Pachube
Correct Answer: C	
Reference: Book name: IoT Hardware And Systems, page no 143	

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Q28: What does a lightning bolt on the top right corner of the Pi screen mean?	
A. Low Processing Power	B. Low Power
C. Low Memory	D. SD Card Low On Memory
Correct Answer: B	
Reference: Book name: IoT Hardware And Systems, page no 114, Heading 8.5	

Q29: Where is the boot config file config.txt stored of the Raspberry Pi?	
A. /pi	B. /boot
C. /root	D. /config
Correct Answer: B	
Reference: Book name: IoT Hardware And Systems, page no 113	

Q30: What is the maximum power that a Raspberry Pi USB port provide?	
A. 110mA	B. 170 mA
C. 140 mA	D. 180 mA
Correct Answer: C	
Reference: Book name: IoT Hardware And Systems, page no 107, Heading 8.2	

Q31: Which of the following options is an invalid variable name in Python.	
A. Testvar2	B. _test_var
C. Test-var	D. testVAR
Correct Answer: C	
Reference: Book name: IoT Hardware And Systems, page no 126	